Year 13 Maths - Normal distribution

Similar questions - Finding mu and sigma, symmetry

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The random variable Y has the distribution $\mathrm{N}(\mu, \sigma^2)$. Given that $\mathrm{P}(Y > 38) = 0.1$ and $\mathrm{P}(Y < 20) = 0.25$, find μ and σ, giving your answers to 3 significant figures. **[6]**

2

(a) $X \sim \mathrm{N}(30, 9)$. Find b such that $\mathrm{P}(30 - b < X < 30 + b) = 0.6$. **[3]**

(b) $W \sim \mathrm{N}\left(\mu, \frac{\mu^2}{16}\right)$. Find $\mathrm{P}(W > 1.25\mu)$. **[3]**

(c) $U \sim \mathrm{N}(8, 4)$ and $V \sim \mathrm{N}(10, 16)$. Given that $\mathrm{P}(U < k) = \mathrm{P}(V > m)$ where $k > 8$ and $m < 10$, find k in terms of m. **[3]**

Answers

Question 1

$\frac{38-\mu}{\sigma} = 1.2816$ and $\frac{20-\mu}{\sigma} = -0.6745$ (from $\Phi^{-1}(0.9)$ and $\Phi^{-1}(0.25)$). Subtracting: $\frac{18}{\sigma} = 1.9560$, so $\sigma = 9.202$ and $\mu = 38 - 1.2816\sigma = 26.207$

Question 2

(a) Symmetry: $\mathrm{P}(X < 30-b) = 0.2$, so $\mathrm{P}(X < 30+b) = 0.8$. $\frac{b}{3} = 0.8416$, $b = 2.52$

(b) $\sigma = \mu/4$. $\mathrm{P}(W > 1.25\mu) = \mathrm{P}\left(Z > \frac{0.25\mu}{\mu/4}\right) = \mathrm{P}(Z > 1) = 0.159$

(c) $\mathrm{P}(U < k) = \mathrm{P}\left(Z < \frac{k-8}{2}\right)$ and $\mathrm{P}(V > m) = \mathrm{P}\left(Z > \frac{m-10}{4}\right) = \mathrm{P}\left(Z < \frac{10-m}{4}\right)$. So $\frac{k-8}{2} = \frac{10-m}{4}$, giving $k = 8 + \frac{10-m}{2} = 13 - \frac{m}{2}$